HALOALKANES & HALOARENES

Class 12 Chemistry | English + Hinglish Revision Notes

How to study / Kaise padhein: Pehle English definition padho, phir Hinglish explanation samjho. Uske baad reaction ko khud likhne ki practice karo.

Important: These notes are for quick revision. Board preparation ke liye NCERT textbook aur current CBSE syllabus bhi check karna.

1. Basic Concepts / Basic Baatein

English: Haloalkanes are compounds in which one or more hydrogen atoms of an alkane are replaced by halogen atoms (F, Cl, Br or I).

Hinglish: Jab alkane ke ek ya zyada hydrogen atoms ki jagah halogen (F, Cl, Br, I) aa jaata hai, toh compound haloalkane kehlata hai.

General formula: R-X. **Hinglish:** Yahan R alkyl group hai aur X halogen ko show karta hai.

English: In haloarenes, the halogen is directly attached to an aromatic ring. Example: chlorobenzene (C6H5Cl).

Hinglish: Haloarene mein halogen seedha benzene jaise aromatic ring se attached hota hai.

Classification / Types

• **Primary (1°):** X wala carbon ek doosre carbon se attached. Example: CH3CH2Cl.

• **Secondary (2°):** X wala carbon do doosre carbons se attached. Example: CH3CHClCH3.

• **Tertiary (3°):** X wala carbon teen doosre carbons se attached. Example: (CH3)3CCl.

IUPAC naming / Naam kaise likhein

• Longest carbon chain choose karo; numbering aisi do ki substituent ko lowest possible number mile.

• Halogen prefixes: fluoro-, chloro-, bromo-, iodo-. Examples: CH3CH2Cl = chloroethane; CH3CHBrCH3 = 2-bromopropane.

2. Preparation of Haloalkanes / Banane ke tareeqe

A. From alcohols

Reaction: R-OH + HX -> R-X + H2O

English: Alcohols react with suitable hydrogen halides or halogenating reagents to form haloalkanes.

Hinglish: Alcohol ko suitable reagent ke saath react karane par -OH group ki jagah halogen aa sakta hai.

B. From alkenes

Reaction: CH2=CH2 + HBr -> CH3CH2Br

English: Hydrogen halide adds across the carbon-carbon double bond.

Hinglish: Alkene ke double bond par H aur Br add hote hain, jisse haloalkane banta hai.

C. Free-radical halogenation

Reaction: CH4 + Cl2 --sunlight/UV--> CH3Cl + HCl

Hinglish: UV light/sunlight mein methane ka chlorine ke saath reaction chloromethane bana sakta hai. Further substitution bhi ho sakti hai.

3. Physical Properties / Physical Gun

• **Polarity:** C-X bond polar hota hai because halogen carbon se zyada electronegative hota hai.

• **Boiling point:** Generally corresponding alkanes se higher hota hai; molecular mass badhne par boiling point often badhta hai.

• **Branching:** Zyada branching generally boiling point ko kam karti hai.

• **Solubility:** Haloalkanes water mein poorly soluble hote hain, but many organic solvents mein dissolve hote hain.

4. Reactions of Haloalkanes / Important Reactions

A. Nucleophilic substitution

English: A nucleophile replaces the halogen atom in a haloalkane.

Hinglish: Nucleophile halogen ki jagah le leta hai.

Reaction: R-X + OH- -> R-OH + X-

Reaction: R-X + CN- -> R-CN + X-

Ammonia can also react with haloalkanes to form amines; actual products depend on conditions and reagent ratio.

B. Elimination

English: Alcoholic KOH and heat can remove HX from a haloalkane to form an alkene.

Hinglish: Alcoholic KOH + heat par H aur X nikal sakte hain aur double bond ban jaata hai.

Reaction: CH3CH2Br + KOH (alcoholic, heat) -> CH2=CH2 + KBr + H2O

Exam tip: Aqueous KOH usually favours substitution; alcoholic KOH with heat usually favours elimination.

C. Wurtz reaction

Reaction: 2R-X + 2Na --dry ether--> R-R + 2NaX

Hinglish: Dry ether mein sodium ke saath haloalkane react karke higher alkane bana sakta hai. Symmetrical alkanes ke liye useful hai.

5. SN1 vs SN2 / Dono mein farq

Feature	SN1	SN2
Steps	Two main stages; carbocation intermediate.	One concerted step; no free carbocation intermediate.
Rate law	Rate = k[R-X]	Rate = k[R-X][Nu-]
Substrate trend	Usually 3° > 2°; 1° generally unfavourable.	Usually methyl > 1° > 2°; 3° strongly hindered.
Stereochemistry	May give racemisation/partial racemisation at a chiral centre.	Backside attack causes inversion at a chiral centre.
Easy memory	SN1: rate depends mainly on haloalkane.	SN2: rate depends on haloalkane and nucleophile.

Hinglish trick: SN1 mein carbocation banta hai aur rate mainly R-X par depend karti hai. SN2 ek hi step mein hota hai aur nucleophile bhi rate ko affect karta hai.

6. Haloarenes / Aromatic Halides

English: Haloarenes contain a halogen directly attached to an aromatic ring, e.g., chlorobenzene (C6H5Cl).

Hinglish: Chlorobenzene mein chlorine directly benzene ring se attached hota hai.

Why are haloarenes less reactive in nucleophilic substitution?

• **Resonance:** C-X bond gets partial double-bond character, so it becomes shorter and stronger.

• **Hinglish:** Resonance ki wajah se C-X bond mein partial double-bond character aa jaata hai, isliye bond todna difficult hota hai.

• The carbon attached to halogen is sp2-hybridised.

• Formation of a phenyl carbocation is highly unfavourable, so the usual SN1 pathway is difficult.

Electrophilic substitution

Halogens deactivate the ring by their -I effect, but direct incoming electrophiles mainly to ortho and para positions due to resonance donation.

Hinglish: Halogen ring ko overall deactivate karta hai, lekin resonance ke kaaran incoming group ko mainly ortho/para positions par direct karta hai.

7. Named Reactions and Tests / Naam wali Reactions

Reaction / Test	English + Hinglish summary
Wurtz	2R-X + 2Na (dry ether) -> R-R + 2NaX. Do alkyl groups join hote hain.
Finkelstein	Alkyl chloride/bromide ko NaI in acetone se alkyl iodide mein convert karne ke liye use kiya jaata hai.
Swarts	Suitable metal fluorides ki help se alkyl fluorides prepare kiye ja sakte hain.
Silver nitrate test	Some haloalkanes alcoholic AgNO3 ke saath AgX precipitate dete hain; result substrate/conditions par depend karta hai.

Silver halide colours: AgCl = white, AgBr = cream/pale yellow, AgI = yellow.

Hinglish: AgCl safed, AgBr cream/pale yellow aur AgI yellow precipitate deta hai.

8. Quick Revision / Jaldi Repeat Karo

Question	Short answer
Aqueous KOH vs alcoholic KOH?	Aqueous KOH: usually substitution. Alcoholic KOH + heat: usually elimination.
Why is aryl C-X bond stronger?	Resonance gives partial double-bond character; bonded carbon is sp2-hybridised.
Why is SN2 difficult for 3° haloalkanes?	Steric hindrance backside attack ko block karti hai.
Why are haloalkanes poorly soluble in water?	Water-water interactions todne ki energy generally hydration se fully compensate nahi hoti.

9. Board-Style Practice / Exam Questions

- Define haloalkanes and haloarenes with one example each. / Dono ki definition aur example likho.
- Write the IUPAC name of CH3CHBrCH3. / Iska IUPAC naam likho.
- Differentiate between SN1 and SN2 using three points. / Teen differences likho.
- Why are haloarenes less reactive towards nucleophilic substitution? / Reason explain karo.
- Write Wurtz reaction and mention one limitation. / Reaction aur ek limitation likho.
- Explain the role of alcoholic KOH. / Alcoholic KOH ka use batao.
- State the colours of AgCl, AgBr and AgI. / Precipitate ke colours yaad karo.

10. Last-Minute Checklist / Final Revision

- Learn classification and IUPAC naming. / Types aur naming practise karo.
- Memorise reaction conditions: aqueous KOH, alcoholic KOH + heat, dry ether, UV light.
- Revise the SN1/SN2 comparison table.
- Learn why haloarenes are less reactive in nucleophilic substitution.
- Practise named reactions and silver halide colours.

Study note: These notes are a concise revision aid, not a replacement for NCERT. Current CBSE syllabus and prescribed textbook ko final authority samjho.